INCH-POUND
MIL-PRF-55365D
3 July 1997
SUPERSEDING
MIL-C-55365C
23 AUGUST 1990

PERFORMANCE SPECIFICATION

CAPACITOR, FIXED, ELECTROLYTIC (TANTALUM), CHIP, NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

FAILURE RATE LEVELS M, P, R, AND S
ARE INACTIVE FOR NEW DESIGN AFTER 23 AUGUST 1990.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for nonestablished reliability (non-ER) and established reliability (ER), tantalum dielectric, fixed chip capacitors, primarily intended for use in thick and thin film hybrid circuits or surface mount applications for filter, bypass, coupling, and other applications where the alternating current (ac) component is small compared to the direct current (dc), rated voltage and where supplemental moisture protection is available (see 6.1). The established reliability capacitors have reliability ratings established on the basis of life tests performed at specified voltage at +85°C for failure rate levels (FRL) ranging from:

a. 1.0 percent per 1,000 hours to 0.001 percent per 1,000 hours in accordance with MIL-STD-690. These FRL's are established at a 60-percent confidence level and are maintained at a 10-percent producer's risk (exponential distribution).

b. 0.1 percent per 1,000 hours to 0.0001 percent per 1,000 hours (1 FIT) at 90-percent confidence level (Weibull distribution). 1/

1/ FIT = failure unit = one failure per 10^9 device hours.

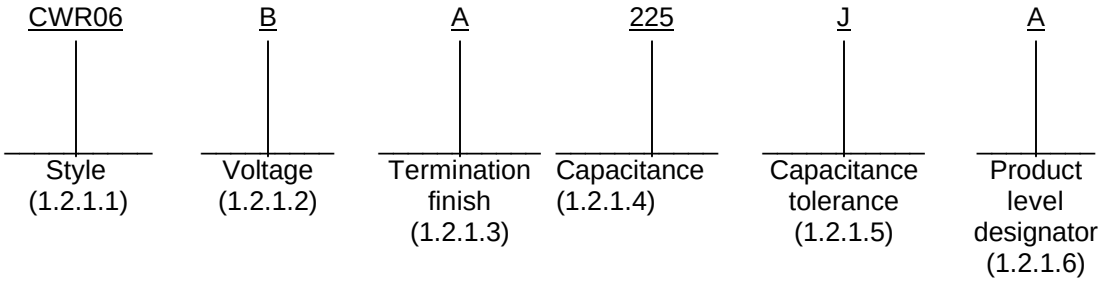
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

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1.2 Classification. Capacitors covered by this specification are classified by style as specified (see 3.1).

1.2.1 Part or Identifying Number (PIN). The PIN should be in the following form and as specified (see 3.1).



1.2.1.1 Style. The style is identified by the three-letter symbol “CWR”, followed by the two digit number. The letters identify tantalum chip capacitors. The number identifies the design of the capacitor.

1.2.1.2 Voltage. The voltage (rated, derated, and surge) is identified by a single letter as shown in table I.

TABLE I. Voltage.

Symbol	Voltage		
	Rated (+85°C)	Derated (+125°C)	Surge (+85°C)
	Volts, dc	Volts, dc	Volts, dc
A	2	1.3	2.6
B	3	2.0	4.0
C	4	2.7	5.0
D	6	4.0	8.0
E	8	5.4	10.0
F	10	7.0	13.0
G	12	8.0	16.0
H	15	10.0	20.0
J	20	13.0	26.0
K	25	17.0	32.0
L	30	20.0	39.0
M	35	23.0	46.0
N	50	33.0	65.0

1.2.1.3 Termination finish. The termination finish is identified by a single letter as follows:

B - Gold plated (50 microinch minimum).
C - Hot solder dipped (60 microinch minimum).
H - Solder plated (100 microinch minimum).
K - Solder fused (60 microinch minimum).

At the option of the manufacturer, all termination finishes may have a barrier metal. See 6.8 for conversion from previous termination finishes.

1.2.1.4 Capacitance. The nominal capacitance value, expressed in picofarads (pF), is identified by a three-digit number. The first two digits represent significant figures and the third digit specifies the number of zeros to follow.

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1.2.1.5 Capacitance tolerance. The capacitance tolerance is identified by a single letter as shown in table II.

TABLE II. Capacitance tolerance.

Symbol	Capacitance tolerance percent (%)
J	5
K	10
M	20

1.2.1.6 Product level designator. The product level designator is identified by a single letter as shown in table III.

TABLE III. Product level designator.

Symbol	Product level
A	non-ER
Exponential FRL (see 1.1)	
	% per 1,000 hours
M	1.0
P	0.1
R	0.01
S	0.001
Weibull FRL (see 1.1)	
	% per 1,000 hours
B	0.1
C	0.01
D	0.001

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-55365/2	-	Capacitors, Chip, Fixed, Tantalum, Established Reliability, Styles CWR03 and CWR04
MIL-PRF-55365/4	-	Capacitors, Chip, Fixed, Tantalum, Established Reliability, Styles CWR06 and CWR09
MIL-PRF-55365/7	-	Capacitors, Chip, Fixed, Tantalum, Established Reliability, Style CWR10
MIL-PRF-55365/8	-	Capacitors, Chip, Fixed, Tantalum, Established Reliability, Style CWR11 (Metric)

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202	-	Electronic and Electrical Component Parts, Test Methods for.
MIL-STD-690	-	Failure Rate Sampling Plans and Procedures.
MIL-STD-790	-	Standard Practice for Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-554-1	-	Assessment of Average Outgoing Quality Levels in Parts Per Million (ppm).
EIA-557	-	Statistical Process Control Systems (DoD adopted).

(Application for copies should be addressed to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Qualification. Capacitors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.4 and 6.4). Authorized distributors which are approved to MIL-STD-790 distributor requirements by the QPL manufacturer are listed in the QPL.

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3.3 QPL system. The manufacturer shall establish and maintain a QPL system for parts covered by this specification. Requirements for this system are specified in MIL-STD-790 and MIL-STD-690. In addition, the manufacturer shall also establish a Statistical Process Control (SPC) and Part Per Million (ppm) system that meets the requirements as detailed in 3.3.1 and 3.3.2 respectively

3.3.1 SPC system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish an SPC system that meets the requirements of EIA-557. Typical manufacturing processes include: Pressing, sintering, electrochemical processing, encapsulating, and packaging.

3.3.2 PPM system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a ppm system of assessing the average outgoing quality of lots in accordance with EIA-554-1. Data exclusion, in accordance with EIA-554-1 may be used with approval of the qualifying activity. The ppm system shall identify the ppm rate at the end of each month and shall be based on a 6-month moving average. Style reporting may include both non-ER and ER style combinations.

3.4 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Interface and physical dimension. Capacitors shall meet the interface and physical dimensions specified (see 3.1).

3.5.1 Body structure. The body structure shall be either conformally coated or molded form (see 3.1).

3.5.2 Terminals. Terminals shall be of a solid conductor, of the dimensions specified (see 3.1), and shall be suitably treated to facilitate soldering.

3.5.2.1 Reprocessing of terminations. The manufacturer (or his authorized category B or category C distributor) may reprocess the terminations of the capacitors supplied to this specification, provided the termination process has been approved by the qualifying activity.

3.5.2.2 Reprocessing option. If the manufacturer (or his authorized category B or category C distributor) reprocesses the terminations of the capacitors as a part of normal production, or as a corrective action for solderability failure, the following shall apply:

a. Following any reprocessing, the electrical measurements as specified in group A, subgroup 1, shall be performed on a 200 piece sample for each 8 hours of manufacturing. If there are one or more defects, the individual inspection lot, or lots, from which the defect originated shall be subjected to 100 percent testing of the electrical measurements of group A, subgroup 1, and shall meet the percent defective allowable requirements as specified in 4.6.1.2.3.1.

b. PPM data following the reprocessing shall be reported every 6 months. The calculation method shall be in accordance with of EIA-554-1.

3.5.2.3 Tin plated finishes. Tin plating is prohibited as a final finish or as an undercoat. Tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent (see 6.10).

3.6 Voltage aging (exponential only). When tested as specified in 4.7.3, capacitors shall meet the following requirements:

DC leakage ..... Shall not exceed the requirement specified in 3.7.
Capacitance ..... Shall be within the tolerance specified (see 3.1)
Dissipation factor ..... Shall not exceed the requirement specified in 3.9.
Equivalent Series Resistance (ESR) ..... Shall not exceed the applicable value specified in 3.13. (when specified, see 3.1)

3.7 DC leakage. When measured as specified in 4.7.4, the dc leakage shall not exceed the applicable value specified (see 3.1).

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3.8 Capacitance. When measured as specified in 4.7.5, the capacitance shall be within the applicable tolerance specified (see 3.1).

3.9 Dissipation factor. When measured as specified in 4.7.6, the dissipation factor shall not exceed the value specified (see 3.1).

3.10 Vibration, high frequency. When capacitors are tested as specified in 4.7.7, there shall be no intermittent contacts of 0.5 ms or greater duration, or arcing or other indication of breakdown, nor shall there be any open-circuiting or short-circuiting or evidence of mechanical damage.

3.11 Thermal shock. When tested as specified in 4.7.8, capacitors shall meet the following requirements:

DC leakage	Shall not exceed the requirement specified in 3.7.
Capacitance	Shall change not more than ±5 percent from the initial measured value.
Dissipation factor	Shall not exceed the requirement specified in 3.9.
ESR (when specified, see 3.1)	Shall not exceed the applicable value specified in 3.13.
Visual examination	There shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking (if applicable).

3.12 Resistance to soldering heat. When tested as specified in 4.7.9, capacitors shall meet the following requirements:

DC leakage	Shall not exceed the requirement specified in 3.7.
Capacitance	Shall change not more than ±5 percent from the initial measured value.
Dissipation factor	Shall not exceed the requirement specified in 3.9.
Visual examination	There shall be no evidence of mechanical damage.

3.13 ESR (when specified, see 3.1). When measured as specified in 4.7.10, the ESR shall not exceed the value specified (see 3.1).

3.14 Moisture resistance. When tested as specified in 4.7.11, capacitors shall meet the following requirements:

DC leakage	Shall not exceed 200 percent of the requirement specified in 3.7.
Capacitance	Shall change not more than ±15 percent from the initial measured value.
Dissipation factor	Shall not exceed 150 percent of the requirement specified in 3.9.
ESR (when specified, see 3.1)	Shall not exceed the applicable value specified in 3.13.
Visual examination	There shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking (if applicable).

3.15 Stability at low and high temperatures. When tested as specified in 4.7.12, capacitors shall meet the following requirements:

Step 1 (+25°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall be within tolerance of the nominal value specified (see 3.1).
Dissipation factor	Shall not exceed the applicable value specified (see 3.1).
ESR (when specified, see 3.1)	Shall not exceed the applicable value specified in 3.13.

Step 2 (-55°C):

Capacitance	Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
Dissipation factor	Shall not exceed the applicable value specified (see 3.1).

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Step 3 (+25°C):

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).
Capacitance ..... Shall change not more than ±5 percent from the step 1 measured value.
Dissipation factor ..... Shall not exceed the requirement specified (see 3.1).

Step 4 (+85°C):

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).
Capacitance ..... Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
Dissipation factor ..... Shall not exceed the requirement specified (see 3.1).

Step 5 (+125°C):

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).
Capacitance ..... Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
Dissipation factor ..... Shall not exceed the requirement specified (see 3.1).

Step 6 (+25°C):

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).
Capacitance ..... Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
Dissipation factor ..... Shall not exceed the requirement specified (see 3.1).
ESR (when specified, see 3.1) ..... Shall not exceed the applicable value specified in 3.13.

3.16 Surge voltage (exponential only, see 3.1). When tested as specified in 4.7.13, capacitors shall meet the following requirements:

DC leakage ..... Shall not exceed the requirement specified in 3.7.
Capacitance ..... Shall change not more than the applicable value specified (see 3.1) from initial measured value.
Dissipation factor ..... Shall not exceed the requirement specified in 3.9.
ESR (when specified, see 3.1) ..... Shall not exceed the applicable value specified in 3.13.

3.17 Life (exponential only, see 3.1). When capacitors are tested as specified in 4.7.14, there shall be no evidence of harmful corrosion or obliteration of marking (if applicable), mechanical damage, intermittent shorts, or permanent shorts or opens.

3.17.1 Qualification inspection. When tested as specified in 4.7.14, capacitors shall meet the following requirements:

At +25°C:

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).
Capacitance ..... Shall change not more than the applicable value specified (see 3.1) from the value obtained when measured as specified in 4.7.5.
Dissipation factor ..... Shall not exceed the applicable value specified (see 3.1).
ESR (when specified, see 3.1) ..... Shall not exceed the applicable value specified (see 3.1).

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At +85°C:

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).

At +125°C:

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).

3.17.2 Conformance and periodic inspection.

3.17.2.1 For group A inspection. When tested as specified in 4.7.3 or 4.7.17, exponential or Weibull as applicable, capacitors shall meet the requirements specified in 3.6 or 3.20. Weibull FRL grading data from the inspection lot that meets the requirements of 3.20 and 4.7.17 shall be accepted in lieu of 3.6 and 4.7.3.

3.17.2.2 For periodic group C life or extended life (see 4.7.14.1). When tested as specified in 4.7.14, capacitors shall meet the following requirements:

At +25°C:

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).
Capacitance ..... Shall change not more than ±10 percent from the value obtained when measured as specified in 4.7.5.
Dissipation factor..... Shall not exceed the applicable value specified (see 3.1).

At +85°C:

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).

At +125°C:

DC leakage ..... Shall not exceed the applicable value specified (see 3.1).

3.18 Solderability. When capacitors are tested as specified in 4.7.15, the dipped portion of the terminations shall conform to the solid-wire termination criteria of method 208 of MIL-STD-202. Solderable surfaces shall be as specified (see 3.1).

3.19 Resistance to solvents. When tested as specified in 4.7.16, marking shall remain legible and shall not smear, and capacitors shall meet the following requirements:

DC leakage ..... Shall not exceed the requirement specified in 3.7.
Capacitance ..... Shall change not more than ±2 percent from the initial measured value.
Dissipation factor ..... Shall not exceed the requirement specified in 3.9.
ESR (when specified, see 3.1)..... Shall not exceed the applicable value specified in 3.13.

3.20 Weibull FRL grading (in lieu of 3.6). When tested as specified in 4.7.17, capacitors shall exhibit decreasing FR with respect to time as evidenced by a value of beta (β) which is less than 0.9; and the instantaneous FR in the last interval shall be no more than the FR specified. After grading, capacitors shall meet the following requirements:

DC leakage ..... Shall not exceed the requirement specified in 3.7.
Capacitance ..... Shall be within the tolerance specified (see 3.1).
Dissipation factor ..... Shall not exceed the requirement specified in 3.9.
ESR (when specified, see 3.1)..... Shall not exceed the applicable value specified in 3.13.

Capacitors tested as specified in 4.7.17 shall be exempt from group A percent defective allowable (PDA) provisions (see 4.6.1.2) and exempt from 3.17.2.2 extended life (see 4.7.14.1).

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3.21 Marking. Molded style capacitors shall be marked in accordance with method I of MIL-STD-1285, and shall be as specified (see 3.1). Polarity marking shall be as specified (see 3.1). All styles shall have the following information marked on the package.

a. "JAN" brand.

b. Rated capacitance.

c. Rated voltage.

d. Capacitance tolerance.

e. FRL symbol.

f. PIN.

g. Manufacturer's source code in accordance with MIL-STD-1285.

h. Lot date code.

3.21.1 "JAN" and "J" marking. The United Stated Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear to certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of the specification and the applicable specification sheets or associated specifications, the manufacturer shall remove completely the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or ""J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 1,586,261 for the certification mark "J".

3.21.2 Substitution of FRL and product levels. The manufacturer may substitute, with procuring agency approval, FRL and product levels in accordance with table IV.

TABLE IV. FRL and product level substitutability.

Parts qualified to FRL	Substitutable for FRL and product level
D	A, M, P, R, S, B, and C
C	A, M, P, R, S, and B
B	A, M, P, R, and S
S	A, M, P, and R
R	A, M and P
P	A, and M
M	A

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3.21.3 Substitution of capacitance tolerance and rated voltage. Parts qualified and marked to tighter capacitance tolerance or higher rated voltage, with procuring agency approval, are substitutable for parts marked to looser capacitance or lower rated voltage, provided all other values, such as case size, characteristic, and terminations remain the same. The substitutable parts shall not be remarked unless specified in the contract or order (see 6.2). In the event the capacitance tolerances or rated voltages are remarked, the lot date codes on the parts shall not be changed and the workmanship criteria shall be met.

3.22 Termination finish code substitutability. Termination finish codes may be substituted for other termination finish codes, with procuring agency approval, in the following manner:

Termination finish code	May be substituted for termination finish code
K	C
C, K	H
C	K

3.23 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.24 Workmanship. Capacitors shall be processed in such a manner that, when examined under 10X magnification, they shall be uniform in quality and shall be free from pits, cracks, rough edges, and other defects that will affect life, serviceability, or function. The capacitors shall exhibit no demetallization (lift-off) on the terminations.

3.24.1 Soldering. All excess flux or solder shall be removed. Electrical connections shall be electrically continuous after soldering.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Conformance inspection (see 4.6).

d. Periodic group C inspection (See 4.6.1.3).

4.2 QPL system. The manufacturer shall establish and maintain a QPL system as described in 3.3. Evidence of such compliance is a prerequisite for qualification and retention of qualification.

4.3 Inspection conditions and methods.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the “GENERAL REQUIREMENTS” of MIL-STD-202.

4.3.2 Methods.

4.3.2.1 AC measurements. AC measurements shall be made at the frequency specified. The magnitude of the ac voltage shall be equal to or less than 1.0 volt root mean square (rms). The maximum dc bias voltage shall be equal to or less than 2.2. volts.

4.3.2.2 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25°C ±5°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

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4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the rated voltage. The power supply employed for dc leakage current measurements shall be stabilized to at least ±100 ppm. During measurements there must be no voltage fluctuations of sufficient amplitude to produce a variation in the current measurement as read with any dc leakage current tester used to test capacitors.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.4) on sample units produced with equipment and procedures normally used in production. Qualification approval will be based on the successful completion of the tests specified in table V, and will not be withheld pending completion of the extended life test of 4.4.4.1.1a.

4.4.1 Sample size. The number and style combinations of capacitors to be subjected to qualification inspection shall be as specified in the appendix to this specification.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table V, in the order shown. All sample units shall be subjected to the inspections of group I. The sample units successfully completing group I inspection shall then be divided as specified in table V for group II through group VI (or VII) inclusive, and subjected to the inspections for their particular group; for combined voltage group submissions, each type shall be equally represented in each group (see 4.6.1.1).

4.4.3 Failures. Failures in excess of those allowed in table V shall be cause for refusal to grant qualification approval.

4.4.4 FRL and quality level verification.

4.4.4.1 FRL qualification.

4.4.4.1.1 Exponential. Exponential FR qualification shall be in accordance with the general and detailed requirements in MIL-STD-690 and the following details:

a. Procedure I: Qualification at the initial FRL. Level “M” (1.0 percent) of FRSP-60 shall apply. Sample units shall be subjected to the qualification inspection specified in group VI, table V (see 4.4.2). The entire life test sample shall be continued on test to 10,000 hours, as specified in 4.7.14.1, on completion of the 2,000 hour qualification tests.

b. Procedure II: Extension of qualification to lower FRL’s. To extend qualification to the “P” FRL, data from two or more voltages within a style may be combined. For FRL’s “R” and “S”, the following styles of similar construction (see 4.6.1.1) may be combined: CWR03 and CWR04; CWR09 and CWR11; CWR06, CWR10, and CWR12.

c. Procedure III: Maintenance of FRL qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification.

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TABLE V. Qualification inspection.

Inspection 1/	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of failures allowed
Group I				
Voltage aging (exponential only, see 3.1)	3.6	4.7.3	178	N/A
DC leakage	3.7	4.7.4		
Capacitance	3.8	4.7.5		
Dissipation factor	3.9	4.7.6		
Equivalent series resistance (when specified, see 3.1)	3.13	4.7.10		
Visual and mechanical examination	3.4, 3.5, 3.21, and 3.24	4.7.2		
Group II				1
Vibration, high frequency	3.10	4.7.7	12	
Thermal shock	3.11	4.7.8		
Group III				
Resistance to soldering heat	3.12	4.7.9	18	
Moisture resistance	3.14	4.7.11		
Group IV				
Stability at low and high temperatures	3.15	4.7.12	12	
Surge voltage (exponential only, see 3.1)	3.16	4.7.13		
Group V				
Life (at +125°C)	3.17	4.7.14	24	
Group VI				
Life (at +85°C)	3.17	4.7.14	102	1
Group VII				
Solderability	3.18	4.7.15	10	0
Resistance to solvents (when specified, see 3.1)	3.19	4.7.16		

1/ For qualification of design changes only, manufacturers may submit Weibull data instead of group V and group VI test data.

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4.4.4.1.2 Weibull. Weibull FR qualification will be granted only to manufacturers who have achieved FRL P for any capacitor style covered by this specification in accordance with 4.4.4.1.1. To extend qualification to include Weibull FRL's, the manufacturer shall demonstrate the capability of Weibull FR grading (see 4.7.17) to the qualifying activity.

If during two consecutive reporting periods there has been no production of the lowest Weibull FRL for which the manufacturer is qualified, the manufacturer may be required, at the discretion of the qualifying activity, to submit a product of each style to testing in accordance with the qualification inspection requirements. Failure to meet this requirement shall result in a loss of the manufacturer's FR to the lowest FR last demonstrated.

4.4.4.2 Quality level verification. The contractor is responsible for establishing a quality system to verify the ppm defect level of lots that are subjected to the group A inspections. The ppm defect level shall be maintained for each specification sheet. The ppm defect level shall be based on a 6-month moving average.

4.5 Verification of qualification. Every 6 months, the manufacturer shall provide verification of qualification to the qualifying activity. Continued qualification is based on meeting the following requirements.

a. MIL-STD-790 program.

b. The capacitor design has not been modified.

c. Lot rejection for group A does not exceed 5 percent or one lot, whichever is greater; not applicable to table VII.

d. Periodic group C inspection.

e. Verification of FRL's.

f. PPM assessment. The ppm level defect shall be maintained for each performance specification sheet.

g. Continued qualification to non-ER (A level) shall be based upon continued maintenance of qualification for the ER part FRL B.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspections.

4.6.1.1 Inspection and production lot.

4.6.1.1.1 Inspection lot (exponential distribution). An inspection lot shall consist of capacitors of the same specification sheet (see 3.1), from the same production line or lines, of the same basic design, produced under essentially the same conditions, and offered for inspection during a single month. Capacitors of the same specification sheet must be maintained to at least the P level. The capacitance values and voltages produced shall be represented in the lot in approximately the ratio of production. Voltage groups shall be as follows:

Group I	2 to 20 volts inclusive
Group II	25 to 50 volts inclusive

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4.6.1.1.2 Inspection lot (Weibull distribution). An inspection lot shall consist of capacitors of the same specification sheet (see 3.1), voltage rating, design, and nominal capacitance rating produced in the same case size. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle. All anodes shall be fabricated from a single identifiable powder lot.

4.6.1.1.3 Production lot. A production lot shall consist of all capacitors of the same style (NOTE: Styles CWR03 and CWR04 may be combined), voltage rating, nominal capacitance value, and termination finish. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle. All anodes shall be fabricated from a single identifiable powder lot. Non-ER and ER lots shall be kept separate.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table VI or table VII, and shall be made on the same set of sample units, in the order shown.

4.6.1.2.1 Non-ER capacitors (A level). The manufacturer shall establish and maintain an inspection system to verify that capacitors meet the dc leakage, capacitance, dissipation factor, ESR and mechanical examination requirements. In-line or process control may be a part of such system. The inspection system shall also include criteria for lot rejection and corrective actions. The inspection system shall be verified under the overall MIL-STD-790 QPL system.

NOTE: Since the non-ER (A level) is the ER design without the mandatory conformance inspection and FRL assessment, the product is still expected to meet the environmental qualification type requirements (e.g., moisture resistance, thermal shock, etc.).

4.6.1.2.2 ER capacitors. Group A inspection shall consist of the inspections specified in table VI and table VII and shall be made on the same set of sample units, in the order shown.

4.6.1.2.3 Subgroup 1 tests

4.6.1.2.3.1 Exponential. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Capacitors failing the tests of subgroup 1 shall be removed from the lot. If during the 100 percent inspection, screening requires that more than 5 percent of the capacitors be discarded due to catastrophic or dc leakage failures, the entire lot shall be rejected.

4.6.1.2.3.2 Weibull. Subgroup 1 tests shall be performed on an inspection lot basis (see 4.6.1.1.2) on 100 percent of the product supplied under this specification. Requirements for the infant mortal period and Weibull FRL grading shall be in accordance with 4.7.17.

4.6.1.2.3.3 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in group A, subgroup 1, as the final step of the manufacturing process, the subgroup 1 test may be eliminated when approved by the qualifying activity. The following criteria shall be complied with:

a. The manufacturer production tests are identical or more stringent than those specified for subgroup 1 tests.

b. One hundred percent of the product shall be subjected to these tests.

c. Failure criteria are identical; the same as, or more stringent than, the subgroup 1 test.

d. Lot rejection criteria are identical to, or more stringent than, the subgroup 1 tests.

e. Once approved, future changes require approval from the qualifying activity.

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TABLE VI . Group A inspection (exponential distribution).

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
Voltage aging (exponential only)	3.6	4.7.3	100% inspection
Subgroup 2			
Mechanical examination (physical dimensions only)	3.5	4.7.2	See table VIII
Subgroup 3			
Visual examination			
Materials	3.4	4.7.2	13 samples 0 failures
Marking	3.21		
Workmanship	3.24		
Subgroup 4			
Stability at low and high temperatures	3.15	4.7.12	13 samples 0 failures
Subgroup 5			
Surge voltage (exponential only)	3.16	4.7.13	13 samples 0 failures
Subgroup 6			
Solderability 1/	3.18	4.7.15	13 samples 0 failures

1/ Not applicable to gold plated termination finishes.

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TABLE VII. Group A inspection (Weibull distribution).

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
Life (accelerated FR) 1/	3.20	4.7.17	100% inspection
Subgroup 2			
Mechanical examination (physical dimensions only)	3..5	4.7.2	See table VIII
Subgroup 3			
Visual examination			
Materials	3.4	4.7.2	13 samples
Marking	3.21		0 failures
Workmanship	3.24		
Subgroup 4			
Stability at low and high temperatures 2/	3.15	4.7.12	13 samples 0 failures
Subgroup 5			
Solderability 3/	3.18	4.7.15	13 samples 0 failures

1/ Exempt from 5 percent PDA; rejects shall not be delivered on the contract or order.
2/ Sampling need only conform to the requirements of 4.6.1.1.1 (exponential distribution) inspection lot.
3/ Not applicable to gold plated termination finishes.

TABLE VIII. Sampling plans for group A, subgroup 2.

Lot size	Sample size
1 - 13	100%
14 - 150	13
151 - 280	20
281 - 500	29
501 - 1,200	34
1,201 - 3,200	42
3,201 - 10,000	50
10,001 - 35,000	60
35,001 - 150,000	74
150,001 - 500,000	90
500,001 - UP	102

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4.6.1.2.4 Subgroup 2 tests .

4.6.1.2.4.1 Sampling plans. Subgroup 2 tests shall be performed on an inspection lot basis. Samples subjected to subgroup 2 shall be selected in accordance with table VIII based on the size of the inspection lot. In the event of one or more failures, the lot shall be rejected.

4.6.1.2.4.2 Rejected lots. The rejected lot shall be segregated from new lots and those lots that have passed inspection. The rejected lot shall be 100 percent inspected by those quality characteristics found defective in the sample. Any defectives found shall be removed from the lot. A new sample of parts shall then be randomly selected in accordance with table VIII. If one or more defects are found in this second sample the lot shall be rejected and shall not be supplied to the specification.

4.6.1.2.5 Subgroups 3 and 4 tests. Subgroup 3 and subgroup 4 shall be performed on an inspection lot basis. Sampling inspection shall be in accordance with table VI or table VII.

4.6.1.2.5.1 Rejected lots. The rejected lot from either subgroup shall be segregated from new lots and those lots which have passed inspection. If a lot is rejected, another 13 samples shall be inspected. If the second sample lot has one or more failures, the entire production lot shall be rejected and shall not be delivered on the contract or order.

4.6.1.2.6 Subgroup 5 tests (exponential only). Subgroup 5 tests shall be performed with 13 sample units from the subgroup 3 or subgroup 4 tests, with no failures allowed.

4.6.1.2.6.1 Rejected lots. The rejected lot shall be segregated from new lots and those lots which have passed inspection. If a lot is rejected, another 13 samples shall be inspected. If the second sample lot has one or more failures, the entire production lot shall be rejected and shall not be delivered on the contract or order.

4.6.1.2.6.2 Disposition of sample units. Sample units which have been subjected to subgroup 5 shall not be delivered on the contract or order.

4.6.1.2.7 Subgroup 5 (Weibull only) or subgroup 6 (exponential only) (solderability).

4.6.1.2.7.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot, as defined in 4.6.1.1.1, and subjected to the solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.2.7.2 Rejected lots. If there are one or more defects, the inspection lot shall be rejected. The manufacturer may use one or more of the following options to rework the lot.

a. The individual production lot, or lots, from which the defect originated shall be individually subjected to the solderability test as required in 4.6.1.2.7.1. Production lots that pass the solderability test are available for shipment. Production lots that fail the solderability test may be reworked only if they are subjected to solder dip procedure in 4.6.1.2.7.2b.

b. The manufacturer shall submit the failed lot to a 100 percent reprocessing of the termination finish in accordance with 3.5.2.1. Thirteen additional samples shall then be selected and subjected to the solderability test with no defects allowed. If the lot fails this solderability test, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

4.6.1.2.7.3 Disposition of samples. The solderability test is considered a destructive test and samples subjected to the solderability test shall not be supplied on the contract.

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4.6.1.2.8 PPM calculations . The manufacturer shall establish a ppm system in accordance with 3.3.2 for assessing and calculating average outgoing quality of capacitors. A ppm rate combining DC leakage, capacitance, dissipation factor, and ESR shall be assessed for lots that have passed the group A inspection. The manufacturer's ppm system shall also address rectification procedures for lots failing ppm assessment. Data from the rectification process shall not be used to calculate ppm.

4.6.1.3 Periodic group C inspection (ER only). Group C inspection shall consist of the tests specified in table IX, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed group A inspection; however, sample units subjected to surge voltage and solderability shall not be used.

4.6.1.3.1 Sampling plan. There shall be 89 sample units of each specification sheet taken from production every 3 months and subdivided as specified for the subgroups listed in table IX and subjected to the tests specified in those subgroups, in the order shown. The maximum and minimum case sizes manufactured during that 3-month period shall be represented in the sample in at least the approximate ratio of production. Allowable failures shall be as specified in table IX.

4.6.1.3.2 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

4.6.1.3.3 Noncompliance. If the sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection that the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7 Methods of inspection and test.

4.7.1 Mounting for testing. Mounting is optional for test environments; however, when specified in the test procedures, the chip capacitors shall be mounted on a suitable substrate (e.g., 96 percent alumina, G30 or FR4 glass epoxy). The substrate material shall be such that it shall not be the cause of, nor contribute to, failure of any test for which it may be used. The capacitors shall be mounted on the substrate as follows:

a. A substrate shall be prepared with metallized surface land areas of proper spacing to permit mounting of chips by soldering the terminations of the chips to the “test card” land areas.

b. Solder paste shall be applied to terminals and substrates as applicable or alternative reflow techniques may be used.

c. The chip shall then be placed across the metallized land areas of the test substrate so as to make contact between chip and substrate land areas.

d. The substrate shall be placed in or on a suitable heat transfer unit at a temperature of +135°C ±15°C for 1.0 minute ±0.1 minute. The substrate shall then be placed upon or enter the +245°C ±5°C hot-plate or tunnel oven. The substrate shall remain on the hot-plate or tunnel oven at +245°C ±5°C until the solder paste melts and reflows forming a homogenous solder bond to the metallized substrate.

e. All excess flux or solder shall be removed.

4.7.2 Visual and mechanical inspection. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.4, 3.5, 3.21, and 3.24).

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TABLE IX. Group C inspection (ER only).

Inspection	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of failures allowed
Subgroup I Thermal shock 1/	3.11	4.7.8	12	1
Subgroup II Resistance to soldering heat Moisture resistance	3.12 3.14	4.7.9 4.7.11	18	
Subgroup III Life (2,000 hours at +125°C) 1/	3.17	4.7.14	24	
Subgroup IV Life (10,000 hours at +85°C) FR (exponential only)	3.17	4.7.14.1	25 minimum per style	See 4.4.4.1
Subgroup V Resistance to solvents 2/	3.19	4.7.16	8	0

1/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, this test, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every 3 years after the deletion as part of long term design verification. If the design, material, construction or processing of the part is changed or, if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirements in case of dispute.

4.7.3 Voltage aging (exponential only) (see 3.6). Capacitors shall be subjected to a minimum of 100 percent of dc rated voltage for 40 hours, minimum, at a temperature of +85°C ±5°C. The voltage aging circuit shall have a series resistance of 3.0 ohms, maximum. Capacitors shall then be stabilized at room temperature and the dc leakage, capacitance, and dissipation factor shall then be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

4.7.4 DC leakage (see 3.7). DC leakage shall be measured using the dc rated voltage ±2 percent at the applicable test temperature (see 3.1), after a maximum electrification period of 5 minutes. A 1,000 ohm resistor shall be placed in series with the capacitor to limit the charging current. A steady source of power, such as a regulated power supply shall be used. Unless otherwise specified (see 3.1), measurement accuracy shall be within ±2 percent or 0.02 microampere (μA), whichever is greater (see 4.3.3).

4.7.5 Capacitance (see 3.8) Capacitors shall be tested in accordance with method 305 of MIL-STD-202. Unless otherwise specified (see 3.1), the following details shall apply:

a. Test frequency: 120 Hz ±5 Hz.

b. Limit of accuracy: Measurement accuracy shall be within ±2 percent of the reading.

c. Magnitude of polarizing voltage: Maximum dc bias shall be 2.2 volts for all ac measurements. The magnitude of the ac voltage shall be limited to 1.0 volt rms.

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4.7.6 Dissipation factor (see 3.9). The dissipation factor shall be measured at a frequency of 120 Hz ±5 Hz (unless otherwise specified, see 3.1) by means of a polarized capacitance bridge. The bridge shall provide a dial reading of 0.1 percent dissipation factor and a measuring accuracy of ± (2 percent of the measured dissipation factor plus 0.1 percent).

4.7.7 Vibration, high frequency (see 3.10). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens: Capacitors shall be mounted on a substrate as specified in 4.7.1.

b. Electrical-load conditions: During the test, the specified dc rated voltage (see 3.1) shall be applied to the capacitors.

c. Test condition letter: D (20g)

d. Duration and direction of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the axis.

e. Measurements during vibration: During the last cycle of each plane, electrical measurements shall be made to determine the intermittent open or short circuits. Intermittent contact and arcing shall also be determined. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms or greater.

f. Measurements after vibration: Not applicable

g. Examination after test: Capacitors shall be visually examined for evidence of mechanical damage.

4.7.8 Thermal shock (see 3.11). Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens: Capacitors shall be mounted on a substrate as specified in 4.7.1.

b. Initial measurements: Capacitance (see 4.7.5).

c. Test condition letter: B.

d. Measurements after thermal shock: DC leakage, capacitance, dissipation factor, and ESR (when specified, see 3.1) shall be measured as specified in 4.7.4, 4.7.5, 4.7.6, and 4.7.10, respectively.

e. Examination after test: Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking (if applicable).

4.7.9 Resistance to soldering heat (see 3.12). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exception shall apply:

a. Mounting of specimens: Capacitors shall be mounted on a substrate as specified in 4.7.1, except the post preheat hot-plate temperatures shall be +260°C ±5°C for a duration of 5 seconds ±0.5 second.

b. Measurements prior to test: DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

c. Test condition letter: C.

d. Measurements after test: After completion of the cleaning process and following a minimum 3-hour cooling period, the dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

e. Examination after test: Capacitors shall be visually examined for evidence of mechanical damage.

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4.7.10 ESR (when specified, see 3.1) (see 3.13). The ESR shall be measured. The following details shall apply:

a. Test temperature and tolerance: +25°C ±5°C.

b. Test frequency: 100kHz ±5 kHz.

c. Limit of accuracy: Measurement accuracy shall be within ±5.0 percent of the reading.

d. Magnitude of polarizing voltage: Unless otherwise specified (see 3.1), the maximum dc bias shall be 2.2 volts for all ac measurements. The magnitude of the ac voltage shall be limited to 0.5 volt rms maximum.

4.7.11 Moisture resistance (see 3.14). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens: Capacitors shall be mounted on a substrate as specified in 4.7.1.

b. Initial measurements: Capacitance as specified in 4.7.5.

c. Number of cycles: 20 continuous cycles except that steps 7a and 7b shall be omitted.

d. Loading voltage: Not applicable.

e. Final measurements: After removal from chamber, capacitors shall be dried for 1 hour at room temperature and, within the next hour, dc leakage, capacitance, dissipation factor, and ESR (when specified, see 3.1) shall be measured as specified in 4.7.4, 4.7.5, 4.7.6, and 4.7.10, respectively.

f. Examination after test: Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking (if applicable).

4.7.12 Stability at low and high temperatures (see 3.15). Capacitors shall be dried at +125°C ±5° C for 30 minutes +4 minutes, -0 minutes, prior to start of test. DC leakage, capacitance, and dissipation factor shall then be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively, at each of the temperatures specified in table X, except that dc leakage measurements at -55°C (step 2) are not required. After the measurements of capacitance and dissipation factor have been made at the -55°C temperature (step 2), rated voltage shall be applied through a 33-ohm resistor for the minimum of 5 minutes. ESR (when applicable, see 3.1) shall be measured at step 1 and step 6 as specified in 4.7.10. The capacitors shall be brought to thermal stability at each temperature.

TABLE X. Temperature for stability test.

Step	Test temperature
	(°C)
1	+25 ±3
2	-55 +0, -6
3	+25 ±3
4	+85 +4, -0
5	+125 +4, -0
6	+25 ±3

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4.7.13 Surge voltage (see 3.16). Capacitors shall be subjected to 1,000 cycles of the applicable surge voltage specified in table I. The ambient temperate during cycling shall be +85°C ±5°C. Each cycle shall consist of 30 seconds +2 seconds, -0 second surge voltage application followed by 30 seconds +2 seconds, -0 second discharge period. Voltage application shall be made through a resistor of 33 ohms. The tolerance of the resistor shall be ±5 percent. Each surge voltage cycle shall be performed in such a manner so that the capacitor is shorted terminal to terminal through a copper bar, or an equivalent low resistance at the end of the 30 seconds +2 seconds, -0 second application. An alternate method of shorting the capacitor is to discharge through the same resistance that is utilized for charging. After the final cycle, the capacitors shall be stabilized at the inspection conditions specified in 4.3, and the dc leakage, capacitance, dissipation factor, and ESR (when specified, see 3.1) shall be measured as specified in 4.7.4, 4.7.5, 4.7.6, and 4.7.10, respectively.

4.7.14 Life (see 3.17). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: As specified in 4.7.1.

b. Test temperature and tolerance:

(1) For qualification: Capacitors being subjected to the test of group V of table V shall be at +125°C +4°C, -0°C. Capacitors being subjected to the test of group VI of table V shall be tested at +85°C +4°C, -0°C.

(2) For group C (2,000 hours proof): Capacitors shall be tested at +125°C +4°C, -0°C.

c. Operating conditions: A minimum of dc rated voltage (see 3.1) or a minimum of derated voltage at +125°C (see table I), as applicable, shall be applied gradually (not to exceed 5 minutes either by a slow buildup of the voltage or through a resistor which shall be shorted out within 5 minutes). Voltage shall be applied continuously, except for measurement periods. The impedance of the voltage source, as seen from the terminals of each capacitor, shall not exceed 3 ohms. Storage batteries or an electronic power supply capable of supplying at least 1 ampere when a capacitor is shorted shall be used.

d. Test condition letter: F (2,000 hours).

e. Measurements during the exposure: DC leakage at the applicable high test temperature shall be measured as specified in 4.7.4 at 0 hour; 240 hours +48 hours, -0 hour; 1,000 hours +48 hours, -0 hour; and 2,000 hours +72 hours, -0 hour.

f. Measurement after exposure: Capacitors shall be returned to the inspection conditions specified in 4.3, and visually examined for evidence of mechanical damage; dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

4.7.14.1 Extended life (exponential only). Capacitors shall be tested as specified in 4.7.14, except the test temperature shall be +85°C +4°C, -0°C, and the duration of the test shall be 10,000 hours. DC leakage shall be measured as specified in 4.7.4 at +85°C at 0 hour; 240 hours +48 hours, -0 hour; 1,000 hours +48 hours, -0 hour; 2,000 hours +72 hours, -0 hour; and every 2,000 hours thereafter until 10,000 hours +96 hours, -0 hour have elapsed. Final measurements shall be in accordance with 4.7.14f.

4.7.15 Solderability (see 3.18). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. Mounting surfaces shall be dipped to cover the normal mounting surfaces. After the test, the solderable surfaces shall be examined.

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4.7.16 Resistance to solvents (when specified, see 3.1) (see 3.19). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following exceptions shall apply:

a. Brushing is not required.

b. Initial measurements: Capacitance (see 4.7.5).

c. Measurements after test: DC leakage, capacitance, dissipation factor, and ESR (when specified, see 3.1) shall be measured as specified in 4.7.4, 4.7.5, 4.7.6, and 4.7.10, respectively.

4.7.17 Weibull FRL grading (see 3.20). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Distance the of heating source from specimens, in inches: Not applicable.

b. Method of mounting: Capacitors shall be mounted by their terminations.

c. Test temperature and tolerance: +85°C +4°C, -0°C.

d. Operating conditions: Accelerated dc voltage, +4 percent, -0 percent, as applicable (see table XI), shall be applied gradually (not to exceed 5 minutes by a slow buildup of the voltage). Maximum nominal acceleration factor (see table XI) shall be 20,000:1. Only the capacitors used in 4.7.17e must be fused. Voltage shall be applied continuously, except for failure count periods. The impedance of the voltage source, as seen from the terminals of each capacitor, shall not exceed 1 ohm. An electronic power supply capable of supplying at least 5 amperes when a capacitor is shorted shall be used. A 1- ampere to 2- ampere fuse shall be connected in series with each capacitor. Slow-blow fuses shall not be used. If separate equipment is used for testing the sample and the rest of the lot, the equipment shall be cross calibrated for temperature, voltage, and time to ensure equivalent test conditions.

e. Minimum sample size for monitoring at the beginning of test prior to infant mortal period: 300 pieces, or 100 percent, whichever is less. If the sample size is less than 100 percent, the remainder shall be subject to the same accelerated dc voltage for the same amount of time.

f. Duration of test: 40 hours minimum. The manufacturer shall record the test start and stop times for each lot tested.

Timing:

Infant mortal period	x_1	x_2	x_3
0.....0	0	0	0
5-15 minutes	See 4.7.17h	40 hours minimum	See 4.7.17k

g. Failure definition: A failure is defined as a blown fuse or equivalent.

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h. Failure count during test: The lot size (see 4.6.1.1.2) to be graded is established after removal of gross defectives (infant mortality) (5 minutes - 15 minutes). The first failure count shall be performed at 2 hours ± 0.3 hour after the test was started. If there are no failures at time x_1, the manufacturer may use one of the following options:

(1) Complete a minimum of 40 hours and compute the failure from MIL-STD-690, table II FRSP-90, based on the number of failures at time x_2, or;

(2) Extend time x_1 from 2 hours to a maximum of 10 hours. A failure cannot be assumed. If there are still no failures, option 4.7.17h(1) shall be used, or;

(3) Make one restart at a higher voltage (if applicable) to induce a failure at time x_1. The manufacturer shall assume no previous hours. The restart voltage and time shall be recorded. If there are still no failures, option 4.7.17h(1) shall be used. If the sample size is less than 100 percent, the remainder of the lot must be subjected to the final determined restart time and voltage.

The number of blown fuses and the time under test shall be recorded to within ±0.1 hour. Calculate the fraction failed, p_1, at time x_1, see 6.7.2, equation 4.

i. Failure count after test: A failure count shall be performed after 40 hours minimum after the test was started. The number of blown fuses and the time under test shall be recorded to within ±0.1 hour. Calculate the cumulative fraction failed, p_2, at time x_2 (see 6.7.2, equation 4). If there are no failures at time x_2, the manufacturer may use one of the following options:

(1) Assume one failure and calculate the cumulative fraction failed, p_2, at time x_2 (see 6.7.2, equation 4), or;

(2) Compute the FR from MIL-STD-690, table II FRSP-90, based on the accelerated part hours generated (see 6.7.2, example C), or;

(3) Continue testing. The start time and stop time shall be recorded. If there are still no failures, option (2) may be used.

j. Lot FR: Determine Z (t) from equation 3 (see 6.7.1). If the desired FR has been achieved, the lot may be removed from test.

k. Continuation grading: If the desired FR has not been reached, the lot may be continued on test. The time to reach the FR goal may be estimated from equation 5 (see 6.7.2). If the time calculated to reach the goal FR is excessive, the lot may be discarded in favor of a new lot. If the lot is continued on test, a new FR shall be performed after the extended test. Calculate the cumulative fraction failed, p_3, at time x_3 (see 6.7.2, equation 4). Determine if the FR has been achieved from 4.7.17j.

l. Measurements after exposure: Capacitors shall be removed from the test, stabilized at room ambient conditions (see 4.3.1) and the dc leakage, capacitance, dissipation factor, and ESR (when specified, see 3.1) shall be measured as specified in 4.7.4, 4.7.5, 4.7.6, and 4.7.10, respectively.

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TABLE XI. Normal acceleration factors.

Grading stress V_a/V_r 1/	Acceleration factor
1.0000	1.0000
1.1000	6.5355
1.2000	42.7128
1.3000	279.1496
1.4000	1,824.3823
1.5000	11,923.2626
1.5276	20,000.0000

1/ V_a = accelerated voltage
V_r = rated voltage

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Tantalum chip capacitors are intended to be used in thin or thick film hybrid circuits or surface mount applications where microcircuitry is indicated.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

d. Title, number, and date of the applicable specification sheet, and the complete type designation (see 3.1).

6.3 Supplying for logistic support. Chip components require use of sophisticated equipment to remove from and install on printed wiring boards. Only requisitioners with in-house or contracted capability to replace surface mounted components should be supplied with chip components, in accordance with their specification.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus (DSCC-VQP), 3990 East Broad Street, Columbus, OH 43216-5000. 1/

6.5 Standard capacitor types. Equipment designers should refer to MIL-STD-198, "Capacitors, Selection and Use of", for standard capacitor types and selected values chosen from this specification. MIL-STD-198 provides a selection of standard capacitors for new equipment design.

6.6 Soldering heat. Caution should be exercised when subjecting these sample units to soldering heat. Preheat and soldering exposure times and temperatures should be held to a minimum.

6.7 Weibull FRL determination. Weibull FRL determination is based on lot by lot, 100 percent quality conformance accelerated FR life testing. For example:

2,500	Capacitors have a voltage rating (V_r) of 50 V dc;
X40	hours Weibull life test as 65 V dc voltage applied (V_a);
X279.1496	accelerated factor for V_a/V_r = 1.3000.
27, 914, 960	Accelerated part hours

Weibull FRL's are determined from actual lot performance data. Exponential FRL determination starts with several production lots which may be included in the same inspection lot. For example, 4 production lots of 2,500 capacitors having a voltage rating of 50 V dc are offered for inspection in the same inspection lot.

10,000	Capacitors have a voltage rating (V_r) of 50 V dc;
X40	hours voltage conditioning at 50 V dc minimum;
400,000	part hours, however, exponential lot voltage conditioning performance data are not used to determine FRL's.

110	Samples are drawn from the inspection lot of 10,000 capacitors;
X2,000	hours group C life test at 50 V dc voltage applied;
220,000	part hours, however, data accumulated and used to determine FRL's.

10	Samples selected on completing each group C inspection;
X9,760	hours continuation life testing to 10,000 hours;
97,600	rated condition part hours for FRL maintenance.

Exponential FRL's are based on the aggregate averages of a few samples drawn from many lots maintained in accordance with MIL-STD-690.

1/ SD-6, Provisions Governing Qualification, is issued for the information of applicants requesting qualification of products. Copies of this publication may be obtained from the Defense Printing Service Detachment Office, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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Time ordered distribution of failures for solid tantalum capacitors is described by the Weibull equation:

Equation 1

$$F(x) = 1 - \exp\left[-\frac{x^{b}}{a}\right]$$

Where F(x) = cumulative fraction failed (p) at time x

x = actual test time

β = Weibull “shape parameter” (beta)

α = Weibull “scale parameter”

This relationship may be plotted on graph paper which is constructed with ln x as abscissae and ln (ln (1/(1-p))) as ordinates. Auxiliary scales allow plotting x and p directly. A straight line is obtained. The slope of this line is β, and the y-intercept is -ln α. Figure 1 illustrates a typical Weibull plot.

At any time x, values for β and p can be obtained and the lot FR Z (x) may be calculated from equation 3. A second plot of FR versus time may be drafted as indicated on figure 2. The slope of this line is β -1. Acceptable capacitor lots always exhibit decreasing FR with respect to time as evidenced by a value of β which is less than unity.

6.7.1 Acceleration factors. In order to provide the equivalent of several thousand hours of testing within a practical time frame, voltage acceleration is employed. It has been determined that the application of voltage in excess of rated voltage produces a higher FR than that observed when the devices are operated at the nominal voltage rating. On the Weibull plot, a straight line, parallel to the line representing rated voltage is obtained. The increased number of failures indicated by the line representing the higher voltage results from increased dielectric stress. The slopes (β) of both lines are essentially the same, but the time (x) required to produce any specified p is reduced as voltage is increased. As a result, acceleration factors may be specified which define the relationship between operation at rated voltage and operation at higher-than-rated voltages. For example, a lot of capacitors having a voltage rating of 50 V dc might be tested at 65 V dc. In this case, the ratio of applied voltage to rated voltage is 1.30, resulting in an acceleration factor (A) of 279. In practical terms, operation of these capacitors for 1 hour at 65 V dc is equivalent to operation at 50 V dc for 279 hours. This relationship may be mathematically represented as:

Equation 2

$$Z(t) = Z(Ax) = \left[\frac{b}{a}\ X^{b-1}\right]\left[\frac{1}{A}\right]$$

In conjunction with equation 1, this function may be restated as:

Equation 3

$$Z(t) = F = \frac{-b \ln (1-p)}{x}\left(\frac{10^{5}}{A}\right)$$

The 10^5 factor allows for expression of Z(t) in terms of percent per 1,000 hours when x denotes hours. Table XI illustrates a range of acceleration factors normally used for Weibull FR determination.

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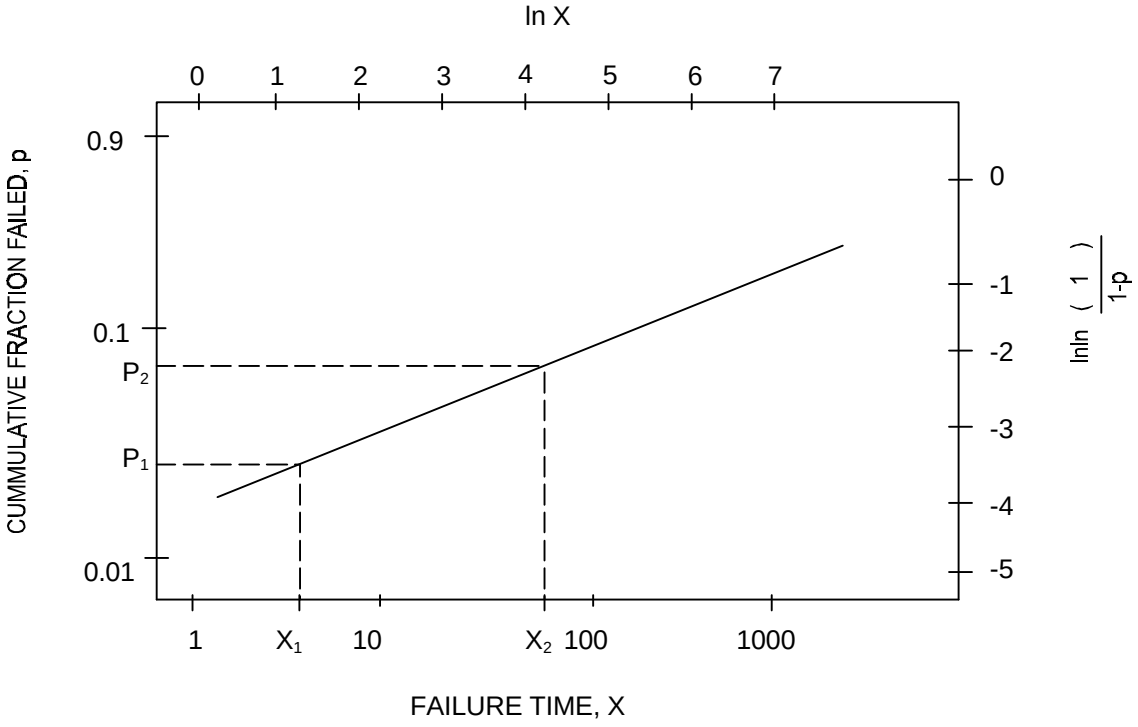


FIGURE 1. Typical weibull plot.

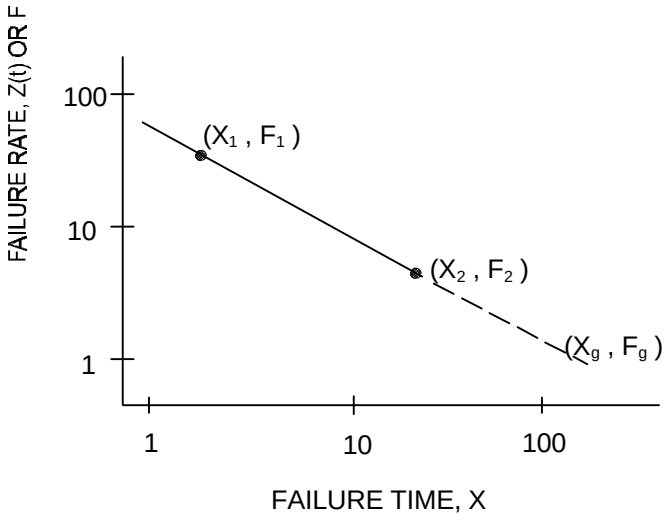


FIGURE 2. Failure rate versus time.

6.7.2 Grading calculations. On the basis of failure counts at x_1 and x_2 as specified in 4.7.17, the slope

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between these points is calculated as follows:

Equation 4

$$b = \frac{\ln\left[\ln\left(\frac{1}{1-p_2}\right)\right] - \ln\left[\ln\left(\frac{1}{1-p_1}\right)\right]}{\ln X_2 - \ln X_1}$$

The FR at time x_2 is then determined from equation 3:

$$F_2 = \frac{-b \ \ln(1-p_2)\ X\ 10^5}{X_2 A}$$

If additional grading time is required to reach the desired FR, the required time x_g may be determined as follows:

Equation 5

$$\ln\ X_g = \frac{\ln F_g - \ln F_2}{b\ -\ 1} + \ln\ X_2$$

Equation 6

$$A = 7.03412025\ X\ 10^{-9}\ e^{(18.77249321\,x\,\frac{V_a}{V_r})}$$

A = Acceleration factor

e = Natural logarithm

V_a = Accelerated voltage

V_r = Rated voltage

Examples:

a. 880 capacitors tested at a grading stress level of 1.2300 (75.0139 acceleration factor) for 40 hours resulted in zero failures.

880 (75.0139 x 40) = 2,640,489 hours

C = 0

FR = B level (MIL-STD-690 FRSP-90)

b. 1,350 capacitors tested at a grading stress level of 1.3300 (490.2535 acceleration factor) for 40 hours resulted in zero failures.

1,350 (490.2535 x 40) = 26,473,689 hours

C = 0

FR = C level (MIL-STD-690 FRSP-90)

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c. 400 capacitors tested at a grading stress level of 1.4000 (1824.3823 acceleration factor) for 40 hours resulted in 1 failure at x_1; no additional failures at x_2.

400 (1824.3823 X 40) = 29,190,117 hours

C = 1

FR = B level (MIL-STD-690 FRSP-90)

d. 100 capacitors tested at a grading stress level of 1.4000 (1824.3823 acceleration factor) for 41 hours resulted in 3 failures at x_1; no additional failures at x_2.

100 (1824.3823 x 41)= 7,479,967.430

C = 3

FR = B level (MIL-STD-690 FRSP-90)

OR assume one additional failure at x_2

x_1 = 2 hours

x_2 = 41 hours

p_1 = .03

p_2 = .04

A =1824.3823

$$b = \frac{\ln\left[\ln\frac{1}{1-p_2}\right] - \ln\left[\ln\frac{1}{1-p_1}\right]}{\ln X_2 - \ln X_1}$$

$$= \frac{\ln\left[\ln\frac{1}{1-.04}\right] - \ln\left[\ln\frac{1}{1-.03}\right]}{\ln 41 - \ln 2}$$

$$= \frac{\ln(\ln 1.041666) - \ln(\ln 1.030928)}{3.713572 - 0.693147}$$

$$= \frac{-3.1985499 - (-3.4913617)}{3.02042425}$$

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$$\frac{0.2928118}{3.02042425} = 0.096944$$

$$FRL = \frac{-\text{b}\ \ln(1 - p_2)\ X\ 10^5}{X_2 A} = \frac{-0.096944\ \ln\ (0.96)\ X\ 10^5}{41\ (1824.3823)}$$

$$= \frac{-0.096944\,(-0.040822)\ X\ 10^5}{74799.67} = 0.000000053\ X\ 10^5$$

$$= \frac{0.53\%}{1000\ \ hours}$$

To compute hours needed to verify 0.001% per 1,000 hours FRL:

$$\ln X_g = \frac{\ln F_g - \ln F_2}{\text{b} - 1} + \ln X_2$$

x_2 = hours at point 2

x_g = hours to test (goal)

F_2 = observed FRL at x_2

F_g = FRL (goal)

$$\ln\ X_g = \frac{\ln(0.001) - \ln\ (0.0053)}{-0.903056} + \ln 41$$

$$= \frac{-6.9077553 - (-5.2400485)}{-0.903056} + 3.713572$$

$$= \frac{-1.6677068}{-0.903056} + 3.713572$$

$$= 1.8467369 + 3.713572 = 5.5503089$$

$$X_g = 259.90\ \ hours$$

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6.7.3 Weibull grading method. After determining the lot FR per 4.7.17, the balance of the lot (when applicable) should be tested to the same voltage acceleration conditions as the monitored test samples. These sample units shall then be subjected to the 100-percent electrical tests shown in table VII.

6.8 Termination finish code conversion. Termination finish codes in this revision (see 1.2.1.3) replace those from earlier revisions in the following manner:

Termination finish codes	Replace earlier revision termination finish codes
B (gold plated)	B (gold)
C (hot solder dipped)	A (solder-coated nickel)
H (solder plated)	C (solder-coated gold)
K (solder fused)	D (solder-coated alloy 725)
	F (solder-coated alloy 752)

6.9 Subject term (key word) listing.

Capacitor
Chip
Established reliability
Tantalum
Weibull

6.10 Tin plated finishes. Tin plating is prohibited (see 3.5.2.3) because it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information, see ASTM B545, “Standard Specification for Electrodeposited Coating of Tin.”

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX

PROCEDURES FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the procedures for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedures for extending qualification of the required sample to other capacitors covered by this specification are also obtained herein. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. SUBMISSION

30.1 Sample.

30.1.1 Single-style submission. A sample of the size required in table V, of the highest capacitance value in each voltage rating in each style for which qualification is sought, shall be submitted.

30.1.2 Combined-voltage submission (exponential). Eighty-nine sample units of the highest capacitance value of the lowest voltage and 89 sample units of the highest capacitance value of the highest voltage in each voltage group for each style for which qualification is sought shall be submitted (see table XII).

TABLE XII. Combined-voltage submission.

Style	PIN 1/	Number of sample units 2/	Rated voltage	Voltage group
CWR03 and CWR04	CWR0-B-107-M	89	3	I
	CWR0-J-226-M	89	20	I
	CWR0-M-106-M	89	35	II
	CWR0-N-335-M	89	50	II
CWR06 and CWR09	CWR0-C-107-M	89	4	I
	CWR0-J-226-M	89	20	I
	CWR0-K-156-M	89	25	II
	CWR0-N-475-M	89	50	II
CWR10	CWR10C-337-M	89	4	I
	CWR10J-476-M	89	20	I
	CWR10K-336-M	89	25	II
	CWR10M-226-M	89	35	II
CWR11	CWR11D-476-M	89	6	I
	CWR11J-156-M	89	20	I
	CWR11K-106-M	89	25	II
	CWR11N-225-M	89	50	II
CWR12	CWR12C-227-M	89	4	I
	CWR12J-476-M	89	20	I
	CWR12K-336-M	89	25	II
	CWR12N-106-M	89	50	II

1/ The complete PIN shall include additional symbols to indicate style, where applicable, termination finish, and capacitance tolerance.

2/ 300 sample units for Weibull (see 30.1.3).

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30.1.3 Combined-voltage submission (Weibull). Three hundred sample units of the highest capacitance value of the lowest voltage and 300 sample units of the highest capacitance value of the highest voltage in each voltage group for each style for which qualification is sought shall be submitted (see table XII). Weibull FRL grading shall be performed in accordance with 3.20 and 4.7.17 instead of voltage aging (exponential only) in group I of table V. DC leakage, capacitance, dissipation factor, and ESR (when specified, see 3.1) shall be performed only once. All other exponential tests in table V shall not apply.

30.2 Certification of material. When submitting samples for qualification, the supplier shall submit certification, in duplicate, that the materials used in their components are in accordance with the applicable specification requirements.

30.3 Description of items. The contractor shall submit a detailed description of the capacitors being submitted for inspection, including body, coating, electrode material, terminations, etc.

40 EXTENT OF QUALIFICATION

40.1 Single-style submission. Capacitance-range qualification will be restricted to values equal to or less than the capacitance value submitted. Capacitance-tolerance qualification will be restricted to tolerances equal to and wider than the tolerance submitted. Voltage rating qualification shall be restricted to those submitted.

40.2 Combined voltage submission. Capacitance range qualification will be restricted to values equal to or less than the capacitance value submitted. Capacitance tolerance qualification will be restricted to tolerances equal to and wider than the tolerance submitted. Voltage rating qualification shall be restricted to those submitted.

40.3 Weibull qualification via similarity. Capacitance range qualification will be restricted to values equal to or less than the capacitance value submitted. Capacitance tolerance qualification will be restricted to tolerances equal to and wider than the tolerance submitted. Voltage rating qualification shall be restricted to those submitted.

40.4 Non-ER capacitors (A level). Qualification of the A (non-ER) level is predicated upon meeting the qualification requirements for the established reliability FRL B.

NOTE: Since the non-ER (A level) is the ER design without the mandatory conformance inspection and FRL assessment, the product is still expected to meet the environmental qualification type requirements (e.g., moisture resistance, thermal shock etc.).

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
NASA - NA

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1935)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-55365D	**2. DOCUMENT DATE (YYMMDD)** 970703

3. DOCUMENT TITLE CAPACITOR, FIXED, ELECTROLYTIC (TANTALUM), CHIP, NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMUNICATIONS-ELECTRONICS COMMAND	b. TELEPHONE *Include Area Code)* (1) Commercial (732) 427-3148 (2) AUTOVON 987-3148
c. ADDRESS *(Include Zip Code)* ATTN: AMSEL-LC-LEO-E-EP FORT MONMOUTH, NJ 07703	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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